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Do listeners devalue AI-generated pop music? Exploring negative biases in listeners' responses to AI-labelled vs human-labelled pop music

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ABSTRACT

The advancement of Artificial Intelligence (AI) in creative domains has sparked discussions regarding how listeners perceive and engage with AI-generated music. This study investigated listeners' emotions, perceptions, and attitudes toward AI-generated versus human-composed pop music. The study hypothesized that listeners would rate music labelled as AI-generated lower in terms of liking, quality, positive emotions, sensorial, imaginal, and experiential responses, as well as need for re-experience and purchase intention, compared to music labelled as human-composed. Participants listened to eight AI-generated pop songs, four labelled as AI-generated and four labelled as human-composed. They then rated each song on various dimensions. To ensure a balanced design, label assignment on composer identity was fully randomized across both participants and songs. Contrary to the hypotheses, the participants rated pop songs labelled as AI-generated more highly in positive emotions, including happiness, interest, awe, and energy, compared to those labelled as human-composed. No significant differences were found between purported composer identity in the remaining dimensions. These results suggest that while the perception of AI authorship does influence listeners, the effects are primarily affective rather than sensorial, imaginal, experiential, or behavioural. Notably, considering that listeners rated pop songs labelled as AI-generated more positively in emotions, the findings imply that AI-generated music may be more readily accepted than previously assumed.

1. Introduction

The integration of Artificial Intelligence (AI) in music creation is transforming how music is composed, produced, and consumed. With deep learning and vast musical datasets, AI can generate compositions closely resembling human-composed music (Burnett et al., 2012). AI can generate melodies and lyrics, as well as tailor compositions to various themes and genres. This advancement has democratized music production by allowing anyone to create songs effortlessly, regardless of musical training. AI-generated music is also gaining traction in advertising, film, and gaming (Pateriya, 2024), highlighting its growing commercial relevance. Despite its capabilities, questions arise about whether listeners can engage with AI music as deeply as human-composed music. This study therefore explores whether people hold inherent biases against AI-generated music, influencing their musical engagement and purchase intention.

1.1. Negative bias toward AI-generated arts

Negative bias toward AI-generated art refers to the tendency to evaluate identical content more negatively when it is labelled as AI-generated rather than human-created. This bias has been observed in several dimensions, such as lower ratings of creativity, authenticity, and perceived effort (Hong et al., 2020; Ragot et al., 2020; Shank et al., 2022). One explanation is the uncanny valley effect, a phenomenon where people may experience discomfort when confronted with something that mimics human qualities but lacks full authenticity (Mori, 1970). In artistic contexts, this discomfort arises not from physical appearance, but from how these AI-generated works imitate human artistic expression without fully capturing its emotional depth. While AI can reproduce surface-level musical features of human compositions such as melody or rhythm, it does so by predicting trends based on statistical patterns (Avdeeff, 2019), rather than drawing from personal experiences or intentionality. This absence of expressive intent may

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limit the emotional depth the music conveys (Cooke, 1959), which is often essential for creating emotionally resonant music. As a result, AI-generated music may sound technically proficient but feel emotionally flat, mechanical, or soulless, which can evoke discomfort or scepticism in listeners (Sturm et al., 2018).

Beyond psychological discomfort, anthropocentrism, the belief in human superiority (Faria & Paez, 2014), also contributes to negative attitudes toward AI-generated arts. Artistic creativity has long been regarded as a uniquely human trait, tied to emotion, consciousness, and personal experience. When AI enters traditionally human domains like music or visual arts, it challenges this belief with the unsettling idea that what was once seen as an exclusively human trait may no longer be so. Millet et al. (2023) argue that AI-generated art is harder to reconcile with anthropocentric views because it encroaches on a domain traditionally associated with human soul, unlike tasks involving analytical or mechanical skills. Supporting this, both Ragot et al. (2020) and Millet et al. (2023) found that simply labelling an artwork as AI-generated led people to devalue its creativity, suggesting that when the creator is non-human, people are biased to view the work as less authentic or meaningful, regardless of its quality. This is because if AI can replicate creativity, it threatens the notion of human uniqueness. Consequently, people downplay the value of AI-generated art pieces to preserve the sense of human distinctiveness and protect the boundary between human and machine. This negative bias may also affect emotional responses, with AI-generated art often perceived as less awe-inspiring than human-made pieces (Millet et al., 2023). More broadly, these studies reflect societal resistance to AI in creative fields, driven by fears that automation may erode the cultural and symbolic significance of human expression (Oh et al., 2017). In this light, dismissing AI-generated creativity becomes not just an aesthetic judgment, but a defence of human identity.

1.2. Weaknesses of past studies and research gaps

Previous research on bias toward AI-generated music has yielded mixed findings, with several failing to detect significant negative biases (Pasquier et al., 2016; Zlatkov et al., 2023). A key limitation in these studies is the repeated presentation of musical excerpts. In studies by Pasquier et al. (2016) and Zlatkov et al. (2023), participants evaluated musical excerpts both before and after being informed of the composer's identity. This approach introduces potential anchoring effects, where initial impressions formed during the first evaluation shape subsequent evaluations (Tversky & Kahneman, 1974), therefore masking any underlying biases.

Another limitation lies in the lack of genre specificity. Pasquier et al. (2016) and Zlatkov et al. (2023) did not specify the genres of music used, yet extrapolated their findings to the broader domain of computational creativity. This generalization is problematic as individuals may perceive and engage with different art forms in distinct ways, and their responses can vary significantly across musical genres. This idea is supported by Shank et al. (2022) as they found that negative biases toward AI-generated music were present in classical music but not in electronic music, underscoring the importance of genre as a moderating factor. Hence, narrowing the focus to a specific form of art and genre is essential to allow for a more precise understanding of perceptual biases toward AI-generated art.

Additionally, previous studies rely on broad, simplistic terms to measure the effects of composer identity (Pasquier et al., 2016; Shank et al., 2022; Zlatkov et al., 2023). For instance, Shank et al. (2022) used general items like "How relaxing or stimulating did you find this piece?" and "How positive or negative are the emotions this piece evokes for you?" to measure emotional responses. Similarly, Pasquier et al. (2016) used broad bipolar rating scales such as "Good–Bad" and "Emotional–Unemotional". These broad, single-item measures fail to capture the complexity of how people experience music. Music engagement is not merely about overall liking or perceived quality. It also encompasses emotional, sensorial, imaginal, and experiential dimensions, with its value rooted in creating an immersive affective experience (Hirschman & Holbrook, 1982). If AI-generated music evokes weaker responses in these dimensions than human-composed music, this could partially explain why listeners show reduced engagement. Furthermore, such gaps in listening experience may also influence downstream behaviours, like desire to re-listen or purchase intention, which are key indicators of commercial success. Without accounting for these emotional, sensorial, imaginal, experiential, and behavioural responses, previous research offers an incomplete picture of the value and potential of AI-generated music.

1.3. Psychological responses to music – emotional, sensorial, imaginal, experiential responses

Emotion plays a central role in music engagement, motivating listeners to seek out, enjoy, and re-engage with music (Lonsdale & North, 2011). According to the emotional contagion framework, listeners unconsciously mimic the emotional expressions conveyed in music, internalizing and experiencing those emotions themselves (Juslin, 2001). Music also triggers autobiographical memories, enhancing personal relevance with the music and emotional intensity (Janata et al., 2007).

However, AI-generated music may struggle to evoke these responses as deeply as human-composed pieces. Due to its lack of expressive intent, AI-generated music may be perceived as less capable of conveying authentic emotion or fostering meaningful connections with listeners (White, 2024). As a result, AI-generated music may be less effective at inducing uplifting emotions like happiness or energizing emotions compared to music composed by human. Moreover, Millet et al. (2023) found that AI-generated artworks were perceived as less awe-inspiring, possibly due to a tendency to undervalue creativity produced by non-human agents. Additionally, because AI compositions are built from statistical patterns, listeners may perceive it as overly formulaic, diminishing their interest and emotional impact.

Beyond emotional reactions, sensorial, imaginal, and experiential responses play a crucial role in music engagement, fostering a deeper and more immersive connection between the listener and the music. These responses enhance both emotional and cognitive engagement, increasing enjoyment and attachment to the music. Sensorial responses, such as rhythmic bodily movements (Lacher & Mizerski, 1994), can heighten pleasure and immersion (Juslin, 2013; Katan, 2016). Imaginal responses involve the mental images evoked by music (Taruffi & Küssner, 2019), which deepen the emotional experience (Presicce & Bailes, 2019) and strengthen personal relevance. Experiential responses refer to a state of deep immersion, where listeners fully engage with the music (Jamieson, 2005). This is driven by shifts in attentional focus toward sensory input and imaginative involvement (Herbert, 2011). These interconnected responses collectively contribute to the need for re-experience, the desire to return to a piece of music to re-live its emotional and sensory richness or to discover new nuances. This emotional gratification reinforces attachment to the music and prolongs enjoyment (Conrad et al., 2018; Margulis, 2014).

Yet again, AI-generated music may fall short in provoking these responses as effectively as human-composed music. While AI can effectively replicate surface-level features like melody or rhythm, it often lacks the subtle emotional inflections and intentionality that make music feel meaningful. For instance, AI may not be able to generate lyrics that authentically mirror human emotions or personal experiences, weakening listeners' emotional identification (Zorkeply & Mahamad, 2022). Therefore, AI-generated music may feel emotionally flat, making it harder for listeners to form vivid mental imagery or become fully absorbed in the experience. This can result in a diminished capacity to elicit vivid mental imagery or a sense of full absorption, ultimately reducing the emotional depth of the experience and the motivation to re-listen.

These emotional and experiential effects are particularly relevant in

the context of pop music, a genre designed to be instantly resonant and emotionally engaging. Pop music often aims to evoke strong, uplifting emotions through catchy melodies and relatable lyrics (Ganser & Huda, 2010). The simplicity and repetition in pop music enhance sensory engagement (Juslin, 2013), by making it easy for listeners to respond to it physically. The emotionally charged nature of pop music also fosters imaginal responses, encouraging listeners to associate the music with personal memories or vivid mental images. Structural features like memorable hooks and choruses are intentionally crafted to capture and maintain attention (Balen et al., 2013), deepening experiential engagement. These qualities make pop music an ideal genre for examining listeners' emotional and experiential responses, and for testing whether AI-generated compositions can evoke reactions comparable to those elicit by human-created music.

1.4. Overview of present study

While existing literature on AI-generated music typically compares it directly to human-composed music (Yin et al., 2023), our study adopts a different approach. Rather than manipulating the musical content, we manipulate perceived authorship by randomly labelling the same set of AI-generated pop songs as either AI-generated or human-composed. This allows us to investigate whether listeners' responses are shaped by the label itself, independent of the actual music. Perceptions are not formed in a vacuum, but are shaped by prior experiences and expectations (de Lange et al., 2018). If people have associated AI-generated music with being less emotional, less immersive, or less meaningful, then labelling a song as AI-composed may activate those expectations and influence their evaluations. In this way, existing literature on how audiences respond to actual AI music informs the cognitive and emotional expectations that listeners bring into our study. By focusing on composer identity labels, our study explores whether perceived authorship alone can influence listeners' emotional, sensorial, imaginal, experiential responses, as well as desire to re-experience and purchase intention.

To address limitations in earlier work, the study employed a within-subject design. Each participant listened to all eight musical excerpts only once while being informed of its purported composer identity. This approach mitigates potential anchoring effects from repeated exposure and allows each participant to evaluate music in two conditions. By controlling for individual differences in musical taste, state mood, and attitudes toward AI, the within-subject design increases the statistical power to detect subtle perceptual biases. The study also goes beyond basic evaluations of liking and quality by assessing a wider range of responses, including emotional (happiness, interest, awe, and energy), sensorial, imaginal, and experiential engagement, as well as the desire to re-listen the music and purchase intention. This approach offers a more comprehensive view on the practical and commercial relevance of audience reactions to AI-generated pop music.

The study specifically focuses on modern pop music, chosen for its wide appeal and accessibility across diverse demographics. All musical excerpts used in the experiment were AI-generated, which reflects the realistic capabilities of AI in music composition. This allows us to assess listeners' responses to music that could feasibly be encountered in real-world settings. To create a listening experience that mirrors contemporary pop music production, the musical excerpts include both lyrics and vocals. This enhances ecological validity by simulating music that listeners might encounter on mainstream platforms.

To ensure a pleasant and emotionally positive listening experience, all songs adhered to a "bright and chill" theme. This thematic choice aligns with music's core functions of inducing pleasure (Schubert, 2009) and serves to minimize potential confounding effects related to emotion aversion or negative emotional responses. Using emotionally positive music reduces the likelihood that participants' reactions are driven by dislike of the music's emotional tone or by negative moods induced by the music itself, rather than by the purported composer identity. However, it is important to note that the findings are specific to this particular musical style and may not generalize to other genres or pieces with different emotional valence.

The study hypothesizes that music labelled as AI-generated will receive lower ratings in perceived liking and quality, elicit fewer positive emotions (i.e., happiness, interest, awe, energy), evoke weaker sensorial, imaginal, and experiential responses, as well as less desire for re-experiencing and lower purchase intention compared to music labelled as human-composed. Participants listened to eight AI-generated musical excerpts, with four randomly labelled as AI-generated and four randomly labelled as human-composed, and subsequently rated each excerpt on various dimensions. The study has been pre-registered at https://osf.io/ga5c7/?view_only=1d8e6d4f1b054b6db860c3b937d156c1.

2. Methods

2.1. Participants

This study employed a within-subject experimental design with two conditions: AI composer and human composer. A priori power analysis for repeated-measures designs was conducted based on a medium effect size ($f = 0.23$, based on Shank et al., 2022), with power set at 0.80 and $\alpha = .05$. The analysis indicated that a minimum sample of 40 participants were required to detect reliable effects. To account for potential exclusions due to demand characteristics and possible underestimation of the effect size, 90 participants were recruited from the university participant pool in exchange for course credit. Of these, 26 participants were excluded for correctly guessing the study's purpose ($n = 9$) or providing inattentive responses ($n = 17$). The final sample consisted of 64 participants (46 females, 18 males; Age: $M = 20.14$, $SD = 1.82$). Among the final sample, 67.19 % ($n = 43$) reported at least one year of instrumental, vocal, or musical composition training, and 82.81 % ($n = 53$) ranked pop music among their top three favourite genres. Descriptive statistics on participants' musical training backgrounds and genre preferences are presented in Tables 1 and 2.

2.2. Musical excerpts

Eight musical excerpts were generated with Suno AI, a generative AI program designed to generate songs with vocals and instrumentation. All songs were created using the prompt "pop genre, happy and chill." To ensure ecological validity, the songs contained lyrics and human vocalizations, with half featuring female vocals and the other half featuring male vocals. To maintain stimulus unfamiliarity, no well-

Table 1
Descriptive statistics of participants' musical training background.

Training Type	*n*	*M*	*SD*
Instrumental Training	40	4.36	4.59
Vocal Training	11	0.52	1.38
Theory/Composition	30	2.30	3.54

Note. n = number of participants who reported receiving musical training in the respective category. Participants indicated the number of years of training received. If they had received more than one type of training concurrently, each type was counted separately.

Table 2
Descriptive statistics of participants' top 3 ranked genre preferences.

Genre	*n*	*M*	*SD*
Pop	53	1.47	0.70
R&B/Soul	38	2.00	0.77
Indie	21	2.05	0.80
Rock	14	2.21	0.70
Hip Hop	13	2.46	0.52
Classical	12	2.67	0.65
Jazz	11	2.55	0.69
Blues	6	2.00	0.89
Electronic	5	2.80	0.45
Country	4	2.50	0.58
Folk	2	2.00	1.41
Reggae	2	2.50	0.71
Ambient	1	1.00	–
Experimental	1	3.00	–

Note. Participants indicated their top three favourite music genres, with 1 = most preferred. *n* = number of participants who ranked the genre among their top three. *M* = mean ranking (1 = most preferred). *SD*s are not reported for genres with fewer than two respondents.

known lyrics or melodies were used. The experimental stimuli[1] consisted of 30-s excerpts from the chorus section of each song.

2.3. Procedure

Each participant completed the study individually in a standard laboratory setting. Before the experiment, participants received a brief task description and provided informed consent. To prevent demand characteristics, participants were informed that the study tested their general attitudes toward music. They were to wear standardized headphones with the volume set to 40 throughout the experiment. As a sound check, participants listened to an audio clip stating the word "banana" and were required to type what they heard into a text box. Only those who passed the sound check could proceed with the experiment.

Participants then listened to eight 30-s excerpts presented in a randomized order. They were informed that four pieces were composed by a human and the other four by an AI, although in fact, all excerpts were AI-generated. Composer identity labels were randomly assigned across participants and excerpts. To enhance ecological validity, fictional composer names were used to reinforce the label manipulation. AI composers included TuneSoft, RoboBeats, SoundGen, and BotStudio, while human composers included names such as Victoria Moore, Christopher Thompson, Anna Jones, Jason Miller, Morgan Walker, Jack Martin, and Vanessa Johnson (Shank et al., 2022). In the human composer condition, composer names were matched to the vocal gender of the excerpts. The distribution of label assignments across participants for each musical excerpt is presented in Table 3.

Participants were allowed to replay each excerpt once if desired. However, they could not proceed to the next page until the full 30-s excerpt had played at least once.

2.4. Measures

After listening to each piece, participants completed a series of questionnaires assessing their responses to each musical excerpt. The measures were presented in the same order as in the experiment.

[1] A pilot test was conducted to assess whether the eight musical excerpts differed from one another. This approach captures the natural variability present in musical excerpts, thereby increasing the ecological validity of the findings. By including diverse musical excerpts, the study aims to ensure that any observed effects or biases are not merely artifacts of a specific song or musical style but instead reflect broader patterns in how listeners perceive AI-generated versus human-generated pop music. Details of the method and results are provided in Supplementary Material.

Table 3
Distribution of composer identity labels across participants.

Musical Excerpt	Labelled as Human-Composed	Labelled as AI-Generated
Music 1	29	35
Music 2	35	29
Music 3	31	33
Music 4	28	36
Music 5	33	31
Music 6	37	27
Music 7	30	34
Music 8	33	31

Music Evaluation. Two items adapted from Shank et al. (2022) measured two different variables: perceived liking and quality of the musical excerpts. Liking was assessed with the item, "How much did you like the music you just heard?" rated on a 7-point Likert scale ranging from −3 (dislike a great deal) to 3 (like a great deal). Quality was measured with the item, "How would you rate the quality of this piece?" rated on a 5-point Likert scale ranging from 1 (very poor) to 5 (excellent).

Emotional Experience. Participants rated their emotional responses to each musical excerpt on four emotions: happiness, interest, awe, and energy. Each emotion was rated on a 5-point Likert scale ranging from 1 (not at all) to 5 (very strong). These emotions were analysed separately.

Sensorial Responses. Two items from Yingling (1962) assessed sensorial responses, including statements such as "I was moving some part of my body (head, foot, hand) in rhythm with the music." Items were rated on a 6-point Likert scale ranging from 1 (strongly disagree) to 6 (strongly agree). The items were averaged and showed good internal consistency (α = .84).

Imaginal Responses. Imaginal responses were measured using three items adapted from Yingling (1962), including statements such as "The song created a picture in my mind." Items were rated on a 6-point Likert scale ranging from 1 (strongly disagree) to 6 (strongly agree). The items were averaged and showed good internal consistency (α = .88).

Experiential Responses. Experiential responses were assessed using five items from the Aesthetic Experiential Processing Scale (Mizerski et al., 1988), such as "I felt as if I were part of the song". Items were rated on a 6-point Likert scale ranging from 1 (strongly disagree) to 6 (strongly agree). The items were averaged and showed excellent internal consistency (α = .91).

Need for Re-experience. Three items from Lacher (1989) measured the desire to re-experience the music, such as "I would enjoy listening to this song again". Items were rated on a 6-point Likert scale ranging from 1 (strongly disagree) to 6 (strongly agree). The items were averaged and showed excellent internal consistency (α = .93).

Purchase Intention. Purchase intention was assessed using one item adapted from Yi (1990): "How likely are you to purchase this song the next time you shop for music?" rated on a 6-point Likert scale ranging from 1 (Very unlikely) to 6 (Very likely).

3. Results

All statistical analyses were conducted using R Version 4.4.2. Outliers falling outside the M ± 1.5 IQR range in each condition were replaced using winsorization. Skewness was assessed using the adjusted Fisher–Pearson coefficient of skewness. Most variables exhibited skewness within ±0.5, which is acceptable for assuming normality (Bulmer, 1979). However, purchase intention demonstrated a high positive skew of 1.17. Consequently, a log transformation was applied, resulting in an adjusted skewness of 0.43.

To examine the effect of composer identity on listeners' music evaluations, a series of linear mixed-effects models were performed for each of the 11 dependent variables using the lmer function from the *lme4* package. These models included fixed effects of composer identity and random intercepts for participants and excerpts to account for individual

differences and variability across excerpts. Composer identity was coded as a categorical predictor with AI = −0.5 and human = 0.5. The results showed that participants rated the music labelled as AI-generated as inducing more happiness, awe, interest, and energizing emotions than music labelled as human-composed (Table 4). However, no significant effects were found for liking, quality, sensorial, imaginal, or experiential responses, need for re-experience, or purchase intention (Table 5).

Additional linear mixed-effects models were conducted to test the potential moderating effects of sequence on the relationship between composer identity and the dependent variables. Sequence refers to the order in which the excerpt was presented within each condition, as the order could influence participants' responses due to factors like fatigue or habituation. The sequence was coded as an ordinal variable representing the position of each musical excerpt within a condition (e.g., the first song presented was coded as 1, the second as 2, and so on). Interaction terms were non-significant across all dependent variables, indicating that the presentation sequence did not moderate the effect of composer identity (Table 6). However, the main effects of the sequence were significant for most dependent variables, except the quality and purchase intention, suggesting a general decline in music ratings over time (Table 7). The main effects of composer identity on happiness, interest, awe, and energizing emotions remained significant after controlling for sequence.

4. Discussion

4.1. Discussion of results

This study investigated whether listeners exhibit biases in their attitudes toward AI-generated pop music. Specifically, we hypothesized that music labelled as AI-generated would receive lower ratings across liking, quality, positive emotions (happiness, interest, awe, energy), sensorial, imaginal, experiential responses, need for re-experience, and purchase intention, compared to music labelled as human-composed. We manipulated the perceived composer identity (AI vs. human) and asked participants to evaluate each song across various dimensions.

Contrary to our expectations, the findings did not support the hypothesis of negative bias toward AI-generated music. There were no significant differences between AI and human conditions in terms of liking, quality, sensorial, imaginal, experiential responses, desire to re-experience or purchase intention. This result is consistent with Tigre Moura and Maw (2021), who similarly found no bias when all musical pieces were AI-generated. One possible explanation is that the AI-labelled songs did not violate listeners' expectations of what AI-generated music would sound like, and therefore did not trigger a sense of mismatch. Shank et al. (2022) also found that negative bias occurred only in classical music but not in electronic music, possibly because electronic music aligns more closely with the aesthetic typically associated with AI. Their findings suggest that listeners may hold pre-existing expectations of what AI can produce, and that negative biases emerge primarily when AI-generated music exceeds those expectations.

Surprisingly, participants rated AI-generated music significantly higher on positive emotions, specifically happiness, interest, awe, and energy after accounting for presentation order. These heightened emotional responses may reflect novelty or curiosity effects, as prior research suggests that people derive pleasure from novel experiences when using AI (Hasan et al., 2020). Participant demographics may also play a role. All participants were university students, a demographic more likely to be familiar with and accepting of AI technologies in everyday life. Younger individuals tend to be more willing to engage with AI technologies and often find the experience more enjoyable than older adults (Barbul & Bojescu, 2023). Compared to earlier studies that reported scepticism or devaluation of AI-generated content, increased exposure and familiarity to AI technologies nowadays may be fostering more open and positive perceptions over time. As AI becomes more embedded in domains such as entertainment and the arts, listeners may be growing more accustomed to AI-generated content and more accepting of its creative potential. For instance, even passive exposure to AI topics is associated with better public perceptions of AI systems (Eurobarometer, 2017). In this context, the absence of negative bias, and even the presence of positive emotional responses, may reflect a broader cultural shift in how AI creativity is perceived.

Notably, these heightened emotional reactions did not translate into significant differences in other experiential dimensions, such as immersion and meaning making. This may be because the positive emotions elicited by AI-generated music were more superficial or transient. Rather than stemming from deep aesthetic or affective engagement, these positive affective reactions may have been primarily driven by curiosity or the novelty of encountering music labelled as AI-generated. According to Marković's (2012) model of aesthetic experience, the process typically begins with the appraisal of an artwork's interestingness, which captures the audience's initial attention. However, for a richer and more enduring aesthetic experience to unfold, such as narrative interpretation and personal relevance, the artwork must provide sufficient expressive and structural depth to sustain ongoing emotional and cognitive engagement (Marković, 2012). It is possible that AI-generated pop music lacked some of these affordances, thereby limiting listeners' engagement beyond the initial appraisal stage. Nevertheless, the absence of negative responses toward AI-labelled music suggests that listeners were open to emotionally engaging with AI-generated music, even if their engagement remained relatively shallow or exploratory. Rather than viewing AI as a threat to human creativity, listeners may have accepted it as a source of positive affective engagement. Rather than rejecting AI as incapable of artistic expression, listeners may be beginning to redefine what counts as meaningful musical experience, showing curiosity and acceptance toward

Table 4
Descriptive statistics on variables of interest.

Outcomes	AI		Human	
	M	*SD*	*M*	*SD*
Liking	5.41	0.91	5.37	0.99
Quality	3.61	0.74	3.59	0.80
Happy	3.36	0.96	3.26	0.95
Interested	3.18	1.07	3.04	1.15
Awe-inspiring	2.58	1.17	2.46	1.05
Energizing	3.26	1.08	3.12	1.15
Sensorial	3.09	1.34	3.01	1.37
Imaginal	3.21	1.30	3.28	1.27
Experiential	2.76	1.06	2.82	1.18
Re-experience	3.33	1.34	3.29	1.39
Purchase Intention [a]	1.95	1.13	2.02	1.21

[a] Descriptive statistics were calculated using the values prior to log transformation.

Table 5
Linear mixed model analysis of composer identity effects on variables of interest.

Outcomes	Fixed Effect of Composer Identity				
	B	SE	*t*	*p*	η_p^2
Liking	−0.05	0.07	−0.75	0.451	0.001
Quality	−0.02	0.05	−0.38	0.705	0.000
Happy	−0.13	0.07	−2.00	0.047[a]	0.009
Interested	−0.16	0.08	−2.04	0.042[a]	0.009
Awe-inspiring	−0.13	0.07	−1.95	0.052	0.008
Energizing	−0.18	0.08	−2.15	0.032[a]	0.010
Sensorial	−0.12	0.09	−1.36	0.175	0.004
Imaginal	0.05	0.09	0.55	0.585	0.001
Experiential	0.05	0.07	0.68	0.497	0.001
Re-experience	−0.05	0.09	−0.59	0.557	0.001
Purchase Intention	0.03	0.03	0.81	0.418	0.001

[a] $p < .05$.

Table 6
Linear mixed model analysis of composer identity and sequence effects on variables of interest.

Outcomes	Fixed Effects										
	Composer Identity					Sequence			Composer Identity [a]Sequence		
	B	SE	*t*	*p*	η_p^2	*F*	*p*	η_p^2	*F*	*p*	η_p^2
Liking	−0.05	0.07	−0.79	0.432	0.001	4.31	0.005[a]	0.029	2.00	0.113	0.014
Quality	−0.02	0.05	−0.40	0.693	0.000	2.55	0.055	0.017	1.44	0.230	0.010
Happy	−0.13	0.06	−2.07	0.039[a]	0.010	10.41	<0.001[a]	0.067	0.34	0.795	0.002
Interested	−0.16	0.08	−2.12	0.034[a]	0.010	8.48	<0.001[a]	0.055	1.12	0.340	0.008
Awe-inspiring	−0.13	0.07	−2.00	0.046[a]	0.009	6.30	<0.001[a]	0.041	0.38	0.766	0.003
Energizing	−0.18	0.08	−2.29	0.023[a]	0.012	16.33	<0.001[a]	0.101	2.54	0.056	0.017
Sensorial	−0.12	0.08	−1.39	0.166	0.004	3.51	0.015[a]	0.024	1.10	0.348	0.008
Imaginal	0.05	0.09	0.56	0.573	0.001	3.67	0.012[a]	0.024	1.76	0.154	0.012
Experiential	0.05	0.07	0.67	0.503	0.001	4.05	0.007[a]	0.027	0.14	0.939	0.001
Re-experience	−0.06	0.09	−0.65	0.517	0.001	11.80	<0.001[a]	0.075	0.52	0.669	0.004
Purchase Intention	0.03	0.03	0.80	0.425	0.001	2.25	0.082	0.015	1.32	0.266	0.009

[a] $p < .05$.

Table 7
Estimated Marginal Means (EMM) of variables of interest in each level of sequence.

Outcomes	Sequence							
	1		2		3		4	
	M	SE	*M*	SE	*M*	SE	*M*	SE
Liking	5.60	0.11	5.36	0.11	5.32	0.11	5.29	0.11
Quality	3.72	0.09	3.59	0.09	3.54	0.09	3.54	0.09
Happy	3.61	0.13	3.20	0.13	3.21	0.13	3.20	0.13
Interested	3.39	0.13	3.07	0.13	3.11	0.13	2.86	0.13
Awe-inspiring	2.77	0.13	2.49	0.13	2.41	0.13	2.42	0.13
Energizing	3.64	0.17	3.14	0.17	2.90	0.17	3.06	0.17
Sensorial	3.29	0.18	2.98	0.18	2.96	0.18	2.94	0.18
Imaginal	3.42	0.15	3.03	0.14	3.32	0.14	3.20	0.15
Experiential	3.00	0.13	2.70	0.13	2.76	0.13	2.71	0.13
Re-experience	3.75	0.15	3.26	0.15	3.17	0.15	3.06	0.15
Purchase Intention [a]	2.08	0.14	2.02	0.14	1.95	0.14	1.87	0.14

Note. EMM refers to the adjusted mean of the dependent variables for each level of the Sequence after accounting for other fixed and random effects.
[a] EMM is calculated using the values prior to log transformation.

AI-generated creativity, even when it evokes less immersive or personally meaningful responses.

4.2. Implications, limitations, and future directions

The absence of negative bias toward AI-generated pop music suggests that human engagement with pop music may be more flexible and context-dependent than previously assumed. While our study did not directly examine these contextual factors, the lack of aversive responses may be shaped by broader influences such as cultural familiarity (Barnes et al., 2024), technological acceptance (Barbul & Bojescu, 2023), and listeners' growing exposure to AI in daily life (Mori, 1970). Rather than being rigidly tied to perceptions of human authorship or authenticity, emotional responses to AI-generated music may be shaped by evolving expectations and growing openness to non-human creative agents. In general, the study sheds light on how younger generations are redefining traditional boundaries between human and AI artistry. As AI becomes embedded in daily life, the psychological processes underlying music appreciation may shift toward greater inclusivity of non-human agency (Hertzmann, 2018).

This shift carries practical implications for musicians, producers, and industry stakeholders. The receptiveness among younger audiences suggests that AI tools can be meaningfully incorporated into music creation as supportive and human-led collaborators that enhance and diversify creative expression. For example, The Velvet Sundown, an AI-assisted band, has amassed over one million monthly listeners on Spotify (Spotify, nd), reflecting growing public acceptance to music created with AI assistance. Rather than viewing AI and human composers as competitors, our findings support a human-driven hybrid model in which generative tools complement, but do not supplant, human creativity. In this model, AI is used to inspire, extend, and enrich the creative process while humans remain in the primary drivers of artistic direction, interpretation, and meaning making. This may empower emerging artists to explore new forms of musical expression while prompting the industry to rethink how creative value is attributed in AI-assisted works. More broadly, the results signal a changing human-machine relationship in creative domains, with possible implications extending to fields such as design, literature, and visual art.

However, several limitations should be considered when interpreting these findings. First, the study did not account for participants' familiarity with the presented musical excerpts, as familiar music tends to be rated more favourably (Tversky & Kahneman, 1974). Although all songs were AI-generated, and we intentionally selected excerpts that did not closely resemble well-known tracks, it is possible that some segments may have included familiar patterns due to the nature of AI training data. This could have subtly influenced participants' responses or confounded the composer identity manipulation. Future studies should incorporate familiarity checks or manipulation validations to better account for this potential confound.

Second, the emotional and expressive range of the musical excerpts may have been limited. The musical excerpts used in this study were intended to reflect how typical AI-generated pop music typically sounds like, and therefore the excerpts may not have captured the emotional depth and expressive nuances present in human compositions. While this enhances ecological validity, it may also constrain our understanding of how listeners respond to a more sophisticated or emotionally rich AI-generated music. Future research should explore how different levels of human-like creativity or expressiveness in AI-generated music affect listeners' perceptions, to better understand the mechanisms behind potential negative biases toward AI-generated art.

Finally, the study sample consisted entirely of university students, which may limit the generalizability of the findings to broader audiences. While students are a relevant demographic, they may not represent the attitudes of older adults, professional musicians, or more traditional music listeners. Future studies should include diverse participant groups to assess whether openness to AI-generated music varies by age, musical background, or professional involvement. Expanding beyond student samples will be essential to fully understand the broader implications of AI in human perception and production.

5. Conclusions

In conclusion, this study found no negative biases toward AI-generated pop music, with participants rating it more favourably on

positive emotional dimensions like happiness, interest, awe, and energizing emotions. These findings suggest that AI-generated pop music may be perceived as a legitimate and emotionally engaging form of artistry, particularly among younger listeners who may be more familiar with and receptive to AI technologies. While limitations such as the potential lack of emotional depth and complexity in the AI-generated music warrant further exploration, the results offer a more nuanced perspective on commonly held assumptions about resistance to AI in creative domains. This may indicate that AI-generated music may be more readily accepted than previously assumed, and that evolving listener attitudes could reshape how creativity is understood in the digital age.

CRediT authorship contribution statement

Suqi Chia: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Andree Hartanto:** Writing – review & editing, Methodology. **Eddie M.W. Tong:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Ethical approval

This study received ethical approval from the National University of Singapore (NUS) Psychology Department Ethics Review Committee (DERC) under Psych-DERC Reference Code: 2024-July-06.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chbah.2025.100217.

Data availability

Cleaned Data, Analysis Code, Stimuli (OSF)

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Update

Computers in Human Behavior: Artificial Humans

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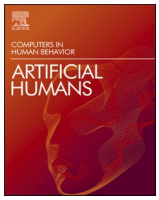
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Erratum regarding corrections related to Ethical approval and informed consent in articles published in [Computers in Human Behavior: Artificial Humans 6C (2025)]

Ethical approval and/or informed consent were not included in the published version of the following articles included in Volume 6C of Computers in Human Behavior: Artificial Humans.

The appropriate Ethical approval and/or informed consent statement, provided by the respective authors, are included below and have been added to the published version of the articles:

1. "Artistic turing test: The challenge of differentiating human and AI-generated art" [CHBAH (2025) 6C: 100209] https://doi.org/10.1016/j.chbah.2025.100209

Ethical approval.

This study was approved by the ethical committee of the Università Campus Bio-Medico di Roma on February 16 of 2022 with number reference number 2021.236.

2. "Exploring the use of generative text AI in design creativity inquiries" [CHBAH (2025) 6C: 100219] https://doi.org/10.1016/j.chbah.2025.100219

Ethical approval.

In Finland, research with human participants must comply with the guidelines of the Finnish National Board on Research Integrity (TENK), The ethical principles of research with human participants and ethical review in the human sciences in Finland. The full details can be found in the Finnish National Board on Research Integrity TENK guidelines 2019.

This study does not contain any of the six factors that trigger the requirement for a formal ethical review and as such has the status of exempt from formal ethical review.

3. "Attractive synthetic voices" [CHBAH (2025) 6C: 100211] https://doi.org/10.1016/j.chbah.2025.100211

Informed consent has been added as a last sentence in the Methods section 2.1 Participants as given below:

Seventy-five participants (39 self-reported as male, 34 self-reported as female, one non-binary, one undisclosed), with an average age of 39.2 years old (SD = 12.1, range: 19–76) were recruited using Prolific (www.prolific.com) and paid an hourly rate of 9£. Inclusion criteria were to be located in the USA, to have English as first language, and to have an approval rate of 95 % or higher in previous studies in the Prolific platform. Informed consent was obtained from all participants prior to data collection.

4. "Nonlinear transformation of probabilities by large language models" [CHBAH (2025) 6C: 100227] https://doi.org/10.1016/j.chbah.2025.100227

Ethical approval declaration section was removed from this article.

5. "Investigating choices regarding the accuracy-transparency trade-off of AI-based systems across contexts" [CHBAH (2025) 6C 100216] https://doi.org/10.1016/j.chbah.2025.100216

The duplicate Ethical approval statement included in section 3.3 Procedures was removed from the article.

6. "Factors influencing users' intention to adopt ChatGPT based on the extended technology acceptance model" [CHBAH (2025) 6C: 100204] https://doi.org/10.1016/j.chbah.2025.100204

Ethical approval and informed consent.

This study was approved by the Institutional Review Board of the University of Dhaka (reference number RECB/DU-MGT/2023/0112). Informed consent was provided by all participants prior to participating in the study through a consent form that clearly stated the study's purpose, confidentiality assurance, and voluntary participation.

7. "Human nature in a virtual world: The attribution of mind perception to avatars" [CHBAH, (2025) 6C: 100222] https://doi.org/10.1016/j.chbah.2025.100222

This study was reviewed and approved by the Ethics Committee of the Rotterdam University of Applied Sciences (REC reference number:

DOIs of original article: https://doi.org/10.1016/j.chbah.2025.100204, https://doi.org/10.1016/j.chbah.2025.100216, https://doi.org/10.1016/j.chbah.2025.100226, https://doi.org/10.1016/j.chbah.2025.100237, https://doi.org/10.1016/j.chbah.2025.100228, https://doi.org/10.1016/j.chbah.2025.100209, https://doi.org/10.1016/j.chbah.2025.100220, https://doi.org/10.1016/j.chbah.2025.100238, https://doi.org/10.1016/j.chbah.2025.100222, https://doi.org/10.1016/j.chbah.2025.100217, https://doi.org/10.1016/j.chbah.2025.100219, https://doi.org/10.1016/j.chbah.2025.100211, https://doi.org/10.1016/j.chbah.2025.100215, https://doi.org/10.1016/j.chbah.2025.100227, https://doi.org/10.1016/j.chbah.2025.100221, https://doi.org/10.1016/j.chbah.2025.100234.

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20231003), and informed consent was obtained from all participants prior to data collection.

8. "The construction and validation of the AI mindset scale (AIMS)" [CHBAH (2025) 6C: 100220] https://doi.org/10.1016/j.chbah.2025.100220

Ethics statement.

Ethical approval for the study was coordinated with the Chair of the Ethics Committee at the Helmut-Schmidt-University/University of the Bundeswehr Hamburg, Germany. In accordance with institutional guidelines, thid study qualified for ethics exemption due to its fully anonymous design and the absence of sensitive or identifiable data. All procedures complied with the ethical principles of the Declaration of Helsinki. All participants provided written informed consent and voluntarily agreed to the processing of their data.

9. "Comparing the audience effect of anthropomorphic robots and humans in economic games" [CHBAH (2025) 6C: 100215] https://doi.org/10.1016/j.chbah.2025.100215

Ethical approval.

Prior to preregistration, this study was assessed by the ethics committee at KTH Royal Institute of Technology, which deemed it exempt from ethical review under Swedish law. The study adhered to Sweden's ethical legislation and relevant guidelines, ensuring compliance with best research practices.

10. "Do listeners devalue AI-generated pop music? Exploring negative biases in listeners' responses to AI-labelled vs human-labelled pop music" [CHBAH (2025) 6C: 100217] https://doi.org/10.1016/j.chbah.2025.100217

Ethical approval.

This study received ethical approval from the National University of Singapore (NUS) Psychology Department Ethics Review Committee (DERC) under Psych-DERC Reference Code: 2024-July-06.

11. "Development and validation of the generative AI engagement scale" [CHBAH (2025) 6C: 100221] https://doi.org/10.1016/j.chbah.2025.100221

Informed consent was added in the Methods section under 2.1 Overview as given below:

All participants received an online information statement and provided informed consent via an online option prior to data collection.

12. "Evaluating the agreement between human preferences, GPT-4V and Gemini Pro Vision assessments: Can AI recognize what people might like?" [CHBAH (2025) 6C: 100234] https://doi.org/10.1016/j.chbah.2025.100234

Informed consent was added in section '2.3. Participants and procedure' as given below:

In addition, we assessed self-reported socioeconomic status (SES), where 8.91 % of participants grouped themselves as below-average, 61.98 % as average and 29.11 % as above-average. Participants provided informed consent prior to taking part in the study. The ethical board of Josip Juraj Strossmayer University of Osijek, School of Applied Mathematics and Informatics approved the study.

13. "Perception of human-robot relationships: Professional, communicative, impersonal and emotionless" [CHBAH (2025) 6C: 100226] https://doi.org/10.1016/j.chbah.2025.100226

Ethical approval and informed consent.

This study was approved by the Ethics Committee of the Leibniz-Institut für Wissensmedien (approval number LEK 2023/040). Informed consent was obtained from all participants prior to data collection.

14. "Why human mistakes hurt more? Emotional responses in human-AI errors" [CHBAH (2025) 6C: 100238] https://doi.org/10.1016/j.chbah.2025.100238

Ethical approval and informed consent.

Participant data were collected through the paid online survey platform Cardemo. All data were collected and stored in an anonymized format, with no personally identifiable information recorded.

This study involved human participants and received ethical review and approval from the Research Ethics Committee (REC) of Hong Kong Baptist University (approval number 2024-25_No. 1 QY). Prior to commencing the experiment, all participants were presented with an online informed consent form that detailed the purpose of the study, the voluntary nature of participation, and their right to withdraw at any time without penalty. Participants were explicitly informed that their responses would remain strictly anonymous and that the collected data would be used solely for academic research purposes. Experimental sessions proceeded only after participants provided digital informed consent.

Research funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

15. "Avatar or human, who is experiencing it? Impact of social interaction in virtual gaming worlds on personal space" [CHBAH (2025) 6C: 100237] https://doi.org/10.1016/j.chbah.2025.100237

Ethical approval and informed consent.

This study was reviewed and approved by the Ethics Committee of the Department of Psychology, Nanjing University (approval ID NJUPSY202302002).

Research Funding statement.

This study was supported by grants from National Natural Science Foundation of China (Grant #31571131) to R.T.

16. "Towards social superintelligence? AI infers diverse psychological traits from text without specific training, outperforming human judges" [CHBAH (2025) 6C: 100228] https://doi.org/10.1016/j.chbah.2025.100228

Informed consent.

All participants provided informed consent prior to completing the study tasks. Consent included permission to use their self-descriptive text in subsequent phases of the research. The consent form described the study purpose, the planned use of their anonymized self-descriptions and questionnaire responses, data handling and anonymization, the fact that participation was voluntary, and their right to decline or withdraw without penalty.